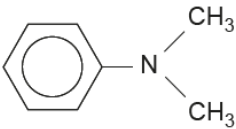


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		award (1) for either of following - sodium nitrite / nitrate(III) / NaNO_2 and HCl / hydrochloric acid - nitrous acid / nitric(III) acid / HNO_2 / HONO	1			1		1
		(ii)				1		1		
		(iii)		$f = \frac{c}{\lambda}$ (1) $7.32 \times 10^{14} \text{ Hz}$ (1)	1	1		2	2	
	(b)	(i)		number of moles = $\frac{0.0075 \times 250}{1000} = 1.875 \times 10^{-3}$ (1) mass required = $272 \times 1.875 \times 10^{-3} = 0.51 \text{ g}$ (1)		2		2	1	
		(ii)		$\frac{1.44}{1.03} = \frac{k \times 0.0096}{k \times c}$ (1) $c = \frac{0.0096 \times 1.03}{1.44} = 6.87 \times 10^{-3} / 0.00687 / 0.0069 \text{ mol dm}^{-3}$ (1) accept alternative method $k = 150$ (1) $c = 6.87 \times 10^{-3} / 0.00687 / 0.0069 \text{ mol dm}^{-3}$ (1)		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		$\text{CH}_3\text{COOH} + \text{NH}_2\text{CONH}_2 \rightarrow \text{CH}_3\text{CONH}_2 + \text{CO}_2 + \text{NH}_3$		1		1		
		(ii)		it contains a <u>nitrogen</u> atom that has a lone pair of electrons / is a proton acceptor		1		1		
		(iii)		the C=O absorption at 1650-1750 cm^{-1} decreases / the N—H absorption at 3300-3500 cm^{-1} decreases (1) the C≡N absorption at 2100-2250 cm^{-1} increases (1)		2		2		
		(iv)		award (1) for any of following - the benzene ring / negative electron cloud is not (easily) susceptible to attack by nucleophiles - the benzene ring / negative electron cloud is usually attacked by electrophiles - the C—Cl bond in chlorobenzene is stronger than the C—Cl aliphatic bond			1	1		
				Question 10 total	2	10	1	13	5	1